

# CREATUSH

## TweenEffects Pro

*Modular DOTween Animation System for Unity*

Technical Documentation | v1.1

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|            |                                  |
|------------|----------------------------------|
| Version    | 1.1.0                            |
| Unity      | 2022.3 LTS or later              |
| Dependency | DOTween (free or Pro)            |
| Namespace  | Creatush.TweenEffectsPro         |
| Assembly   | Creatush.TweenEffectsPro.Runtime |

**TweenEffects Pro is a modular, data-driven animation system for Unity built on DOTween. It replaces the Animator for UI and gameplay animations — no state machines, no clips, no overhead.**

## Architecture

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- VFXBehaviour — abstract base class. Holds timing, ease, loop config, and shared helpers.
- Effects — concrete VFXBehaviour subclasses. Each does one focused thing.
- Controllers — orchestrate one or more effects via EffectControllerBase.
- Support layer — EffectCatalogue, TweenSettingsSO, SplinePathSO.

## Assembly Structure

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| Assembly                         | Contents                                                         |
|----------------------------------|------------------------------------------------------------------|
| Creatush.TweenEffectsPro.Runtime | All runtime code — effects, controllers, base classes, SOs       |
| Creatush.TweenEffectsPro.Editor  | Custom editors, drawers, SplinePathEditor, EffectRewardFlyEditor |

**i** Place the Runtime .asmdef at Runtime/ root. Place the Editor .asmdef at Editor/ root. Do not nest .asmdef files inside subfolders. Editor files (anything using UnityEditor) must live inside Editor/.

## 2. Setup

### Requirements

- Unity 2022.3 LTS or later
- DOTween (free) or DOTween Pro — import before TweenEffects Pro

### Folder Structure

```
Assets/Creatush/TweenEffectsPro/  
  Runtime/                                <- Creatush.TweenEffectsPro.Runtime.asmdef  
    Base/                                VFXBehaviour, EaseCurves, Attributes  
    Effects/                             All VFXBehaviour subclasses  
    Controllers/                         SingleEffect, Sequence, Parallel, Stagger controllers  
    SOs/                                 EffectCatalogue, TweenSettingsSO  
  Editor/                                <- Creatush.TweenEffectsPro.Editor.asmdef  
    VFXBehaviourEditor.cs  
    BaseControllerEditor.cs  
    CurvePresetDrawer.cs  
    CatalogueKeyDrawer.cs  
    SplinePathEditor.cs  
    EffectRewardFlyEditor.cs  
Assets/Resources/  
  TweenSettings.asset                    <- must be named exactly TweenSettings
```

### Global Settings (TweenSettingsSO)

Create via: Assets > Create > Creatush > TweenEffects Pro > Global Settings. Place in any Resources/ folder named exactly TweenSettings. Applied automatically at startup — no scene setup needed.

| Field             | Default | Description                                                        |
|-------------------|---------|--------------------------------------------------------------------|
| globalTimeScale   | 1.0     | Multiplies DOTween.timeScale at startup.                           |
| useUnscaledTime   | false   | Run animations independent of Time.timeScale. Use for pause menus. |
| tweenersCapacity  | 200     | DOTween tween pool size. Increase to avoid runtime pool expansion. |
| sequencesCapacity | 50      | DOTween sequence pool size.                                        |
| safeMode          | true    | DOTween safe mode — catches tween errors instead of throwing.      |
| logWarnings       | true    | Log setup warnings. Stripped in release builds.                    |

## 3. VFXBehaviour — Base Class

Abstract base for every effect. Provides timing, ease, loop, and leak prevention. Subclasses implement `BuildSequence()` and `GetDuration()`.

### Abstract Members

| Member                                           | Description                                                                                      |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <code>BuildSequence(index, total, target)</code> | Build and return a DOTween Sequence. Always end with return <code>FinaliseSequence(seq)</code> . |
| <code>GetDuration()</code>                       | Return wall-clock seconds for one play. Must not allocate. Used by controllers.                  |

### Shared Fields

| Field                     | Default   | Description                                                                     |
|---------------------------|-----------|---------------------------------------------------------------------------------|
| <code>duration</code>     | 0.5s      | Length of one play in seconds.                                                  |
| <code>easeMode</code>     | Preset    | Preset = DOTween Ease enum. Curve = custom AnimationCurve.                      |
| <code>ease</code>         | OutQuad   | Active when <code>easeMode</code> = Preset.                                     |
| <code>easeCurve</code>    | EaseInOut | Active when <code>easeMode</code> = Curve. [CurvePreset] attribute adds picker. |
| <code>loop</code>         | false     | Repeat after completion.                                                        |
| <code>loopCount</code>    | -1        | Repetition count. -1 = infinite.                                                |
| <code>loopInterval</code> | 0s        | Gap between end of one loop and start of the next.                              |

### Protected Helpers

| Method                                      | Usage                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>ApplyEase(Tweener t)</code>           | Apply ease config. Returns Tweener for chaining: <code>seq.Append(ApplyEase(target.DOScale(...)))</code> . |
| <code>FinaliseSequence(Sequence seq)</code> | Apply loop, unscaled time, and <code>SetLink(KillOnDisable)</code> . Always the last call before return.   |

## Leak Prevention

FinaliseSequence() calls SetLink(gameObject, LinkBehaviour.KillOnDisable) — DOTween kills the sequence automatically when the object is disabled. OnDisable() adds a belt-and-suspenders DOTween.Kill(gameObject) for any sequences started outside the normal path.

## Creating a Custom Effect

```
public class EffectMyEffect : VFXBehaviour
{
    [SerializeField] private float myParam = 1f;
    public override float GetDuration() { return duration; }
    public override Sequence BuildSequence(int index, int totalCount, Transform target)
    {
        if (target == null) return null;
        Sequence seq = DOTween.Sequence();
        seq.Append(ApplyEase(target.DOScale(myParam, duration)));
        return FinaliseSequence(seq);
    }
}
```

## 4. Effects

All effects inherit VFXBehaviour and support the shared ease, loop, and timing fields. Every effect calls FinaliseSequence() before returning.

### EffectScalePop

Unified scale effect. Mode field selects the pattern.

| Mode    | Behaviour                                           |
|---------|-----------------------------------------------------|
| ScaleIn | Animate from startScale to endScale once.           |
| Pulse   | Animate to endScale then back. Loops naturally.     |
| Pop     | ScaleIn with elastic overshoot. Springy entry feel. |

### EffectBounce

Position bounce with physically correct squash and stretch. Sequential phases: launch stretch (bottom pivot simulation via position offset) → rise → apex unstretch → fall → landing squash → recovery spring. squashStrength (0–1) controls deformation. bounciness (0–2) controls the OutBack recovery overshoot.

## EffectShake

---

Simultaneous shake on position, rotation, and/or scale — each independently toggleable. rotationStrength is a Vector3 for per-axis control (Z-only for 2D screen shake). vibrato controls oscillation count, randomness controls directional spread, fadeOut smoothly decays the shake. State is restored in OnComplete.

## EffectPunch

---

Elastic punch on any combination of position, rotation, and scale. All active axes run simultaneously via seq.Join. ApplyEase is not applied — DOTween's punch algorithm owns the decay curve. Shape the feel via vibrato and elasticity.

## EffectMove

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Local position from startPosition to endPosition. Snaps to startPosition at play time so the prefab can be in its idle state in the editor.

## EffectMoveTransform

---

A-to-B move using two RectTransform references as waypoints. Drag any two RectTransforms into Origin and Destination. Automatically uses anchoredPosition for UI targets.

## EffectRotate

---

Local rotation from startRotation to endRotation. RotateMode: Full360 = shortest path, FastBeyond360 = continuous spin (default), LocalAxisAdd = adds per loop.

## EffectFloat

---

Sine-wave idle float. Loops by default with InOutSine ease. Optional tiltDegrees adds organic lean. GetDuration returns duration\*2 (full up-down cycle).

## EffectFade

---

CanvasGroup alpha from startAlpha to endAlpha. Requires CanvasGroup (enforced via RequireComponent).

## EffectColorTint

---

Graphic.color from startColor to endColor. Works with any Unity UI Graphic. Requires Graphic component.

## EffectFillAmount

---

Image.fillAmount from startFill to endFill. Requires Image set to Filled type. Logs actionable warning if type is not Filled.

## EffectPivotOrbit

---

Orbit a target around a pivot offset. Temporary pivot GameObject created at runtime, cleaned up in both OnComplete and OnKill.

## EffectSpriteSwap

---

Frame-by-frame sprite animation on a UI Image. Frames array, duration divided by frame count gives per-frame interval. Supports reverse playback. No Animator required.

## EffectSpriteAnimation

---

Full-featured sprite sheet VFXBehaviour on a UI Image (RequireComponent enforced).

| Field / Feature     | Description                                                                 |
|---------------------|-----------------------------------------------------------------------------|
| frames              | Ordered sprite frames array.                                                |
| frameRate           | Frames per second.                                                          |
| playbackMode        | Forward, Reverse, or PingPong (DOTween Yoyo).                               |
| speed               | Playback speed multiplier (0.1–8). Also settable at runtime via SetSpeed(). |
| holdLastFrame       | Keep last frame displayed after non-looping animation ends.                 |
| onAnimationComplete | Fired at the end of each complete pass through all frames.                  |
| onLoopEnd           | Fired at end of each loop iteration (finite loops only).                    |

## EffectFollowSpline

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Moves a target along a cubic Bezier spline defined by a SplinePath component. Features: arc-length remapping for constant speed, orient-to-path with 2D (Z-axis) and 3D (LookRotation) modes, scale-over-path AnimationCurve, roll interpolation between knots, Loop and PingPong playback.

**i** *EffectFollowSpline uses SetLink(KillOnDestroy) instead of KillOnDisable to prevent the sequence being killed when the source GO is disabled mid-animation.*

## 5. Effect Reward Fly

Standalone reward collection effect. Not a VFXBehaviour — manages its own pool, coroutine, and per-item sequences. Trigger via Play() or wire to a UnityEvent.

### Setup

- Add EffectRewardFly to any GameObject inside a Canvas.
- Assign Spawn Parent — a RectTransform inside the Canvas that will parent the pooled items.
- Assign Destination — the RectTransform items fly toward (score counter, wallet icon, etc.).
- Assign Item Prefab — a UI Image with your coin/gem sprite.
- Call Play() from a button, sequence callback, or UnityEvent.

### Fields

| Field                                          | Description                                                   |
|------------------------------------------------|---------------------------------------------------------------|
| itemPrefab                                     | Pooled item prefab. UI Image recommended.                     |
| itemCount                                      | Pool size and number of items per play.                       |
| spawnParent                                    | RectTransform parent for pool items. Must be inside a Canvas. |
| spriteFrames /<br>spriteFrameRate              | Optional sprite sheet animation while item is visible.        |
| spawnRadius /<br>spawnRandomness               | Scatter radius and cluster tightness around the source.       |
| spawnStagger                                   | Seconds between successive item spawns.                       |
| popInOnSpawn /<br>popInDuration                | Scale items from 0 on spawn with OutBack ease.                |
| floatDuration /<br>floatHeight /<br>floatSpeed | Sine-wave idle oscillation before flying.                     |
| destination                                    | RectTransform items fly toward.                               |
| flyDuration / flyEase                          | Fly animation duration and ease.                              |
| arcHeight /<br>arcVariance                     | Parabolic arc peak height and per-item random variation.      |
| flyArrivalScale                                | Scale on arrival. 0 = shrink to nothing.                      |
| flyStagger                                     | Delay between items starting their fly.                       |
| onAllArrived /<br>onItemArrived                | Events fired on each arrival and when all items have arrived. |



## Scene View Gizmo

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The EffectRewardFlyEditor draws a live preview in the Scene View using Handles.matrix set to the spawn parent's local coordinate system. Shows: yellow spawn radius disc, green destination dot, purple float range line, blue arc preview (solid = mean, faded = mean  $\pm$  variance). All gizmo drawing is wrapped in try/finally to guarantee Handles.matrix is always restored.

## 6. Controllers

All controllers extend `EffectControllerBase` which provides shared auto-play, loop, and out-animation logic. Subclasses implement `PlayInternal()` and `GetDuration()` only.

### EffectControllerBase — Shared API

| Method                          | Description                                                                                                               |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>Play()</code>             | Build and play all effects forward.                                                                                       |
| <code>PlayReverse(speed)</code> | Play reversed at the given speed multiplier.                                                                              |
| <code>Hide()</code>             | Play the out animation then deactivate. Immediate <code>SetActive(false)</code> if <code>outAnim.enabled</code> is false. |
| <code>Stop()</code>             | Kill all running tweens owned by this controller.                                                                         |
| <code>GetDuration()</code>      | Wall-clock duration of one play. Uses <code>GetDuration()</code> on each <code>VFXBehaviour</code> — no allocation.       |

### ControllerAutoPlay

| Field                     | Description                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------|
| <code>onStart</code>      | Play once on <code>Start()</code> — first activation only. Skipped if <code>onEnable</code> is also on. |
| <code>onEnable</code>     | Play every time the <code>GameObject</code> is enabled.                                                 |
| <code>loop</code>         | Repeat on a fixed interval while the object is active.                                                  |
| <code>loopCount</code>    | -1 = infinite. Finite count stops the loop coroutine cleanly.                                           |
| <code>loopInterval</code> | Seconds between end of one play and start of the next.                                                  |

### OutAnimSettings

The `outAnim` block configures the exit animation played by `Hide()`. Wire `OnOutComplete` to deactivate parents, trigger transitions, or notify managers.

| Field                      | Description                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------|
| <code>enabled</code>       | Enable out animation. If false, <code>Hide()</code> calls <code>SetActive(false)</code> immediately. |
| <code>reverse</code>       | Play the sequence reversed as the exit animation.                                                    |
| <code>speed</code>         | Playback speed multiplier.                                                                           |
| <code>OnOutComplete</code> | <code>UnityEvent</code> fired when the out animation finishes.                                       |

## SingleEffectController

---

Triggers a single VFXBehaviour on a target transform. playOnEnable and playOnStart supported. Public Play() and Stop() methods.

## SequenceEffectsController

---

Runs effects in sequence. Each step has delay and joinPrevious fields. The Inspector Timeline Preview panel visualises all steps as coloured bars (blue = sequential, green = joined, yellow = delay). GetDuration() sums non-joined steps via VFXBehaviour.GetDuration() — no allocation.

## ParallelEffectsController

---

All steps fire simultaneously via master.Insert(step.delay, stepSeq). Per-step delay offset for staggering within a burst. GetDuration() returns the maximum end time across all steps.

## StaggerEffectsController

---

Runs a single VFXBehaviour on a list of targets with rolling delay. Targets can be explicit or auto-populated from direct children. Supports reverseOrder, randomOrder (Fisher-Yates), and onAllComplete event. index and totalCount are passed correctly to BuildSequence for stagger-aware effects.

# 7. Spline Path System

## SplinePathSO

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Pure data and math asset. Cubic Bezier knots with position, tangentOut, tangentIn, and roll. Binary-search arc-length table for constant-speed remapping. All fields HideInInspector — edited via SplinePathEditor Scene View.

```
// Create via: Assets > Create > Creatush > TweenEffects Pro > Spline Path
```

## SplinePath

---

Scene component. Holds a SplinePathSO reference. Multiple GameObjects can share one asset. Local fallback created automatically if no asset assigned.

## SplinePathEditor

---

Scene View editor for SplinePath. Features: yellow sphere knot handles (click to select), full PositionHandle on selected knot, FreeMoveHandle tangent handles with dotted connecting lines, Add Knot button, Smooth Curves button (Catmull-Rom), per-knot foldout in Inspector, playback override settings. All edits are Undo-wrapped.

## 8. Ease System

Every VFXBehaviour supports two ease modes selectable per effect. EaseMode.Preset uses DOTween's Ease enum. EaseMode.Curve uses a custom AnimationCurve with full tangent control.

### EaseCurves Library

14 hand-tuned AnimationCurve presets. The [CurvePreset] attribute adds a preset picker and Apply button to any AnimationCurve field in the Inspector.

| Group     | Presets                             |
|-----------|-------------------------------------|
| Standard  | SmoothStep, EaseOut, EaseIn, Linear |
| Overshoot | BackOut, BackIn, BackInOut          |
| Bounce    | BounceOut, BounceIn                 |
| Elastic   | ElasticOut, ElasticIn               |
| Specialty | Snap, Spring, Anticipate            |

## 9. Scriptable Objects

### EffectCatalogue

Named library of VFXBehaviour assets. Lazy-builds a lookup dictionary on first Get() call. OnValidate invalidates the cache in the editor.

```
[SerializeField] private EffectCatalogue catalogue;  
[CatalogueKey(nameof(catalogue))]  
[SerializeField] private string effectKey;  
// Runtime:  
catalogue.Play("CoinPop", targetTransform);
```

### TweenSettingsSO

Global project settings. Applied by TweenSettingsLoader at BeforeSceneLoad — no GameObject required. Must be in Resources/ and named TweenSettings.

## 10. Quick Reference

### Effect Selection

| Goal              | Effect                                  |
|-------------------|-----------------------------------------|
| Scale entry       | EffectScalePop (ScaleIn mode)           |
| Idle scale pulse  | EffectScalePop (Pulse mode, loop=true)  |
| Springy pop       | EffectScalePop (Pop mode)               |
| Impact feedback   | EffectPunch (punchScale=true)           |
| Shake/damage      | EffectShake                             |
| Physical bounce   | EffectBounce                            |
| Panel slide       | EffectMove or EffectMoveTransform       |
| Idle float        | EffectFloat (loops by default)          |
| Continuous spin   | EffectRotate (FastBeyond360, loop=true) |
| Fade in/out       | EffectFade                              |
| Color flash       | EffectColorTint                         |
| Health bar        | EffectFillAmount                        |
| Sprite animation  | EffectSpriteAnimation                   |
| Sprite swap       | EffectSpriteSwap                        |
| Path movement     | EffectFollowSpline                      |
| Orbit             | EffectPivotOrbit                        |
| Reward collection | EffectRewardFly                         |

### Controller Selection

| Goal                   | Controller                   |
|------------------------|------------------------------|
| One effect, one object | SingleEffectController       |
| Chain effects          | SequenceEffectsController    |
| All effects at once    | ParallelEffectsController    |
| Same effect on many    | StaggerEffectsController     |
| Exit animation         | Any controller — call Hide() |